



Potential risk factors associated with feline behaviour problems

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ABSTRACT

A retrospective study was carried out on feline behaviour problems presented at the Animal Behaviour Clinic at the Barcelona School of Veterinary Medicine to identify the main risk factors. Three hundred thirty six cats presented for a behaviour problem between 1998 and 2006 were included in the study group. A total of 189 presented at the Hospital of the Barcelona School of Veterinary Medicine for problems other than behavioural and having no record of behaviour problems were used as control group. The main owner's complaint was aggression (47%) followed by inappropriate elimination (39%). 64% of aggression cases involved conflicts between cats and 36% of cases were aggression towards people, owners being the most common target of aggression (78% of all cases of aggression were directed towards people). Play-related aggression and petting-related aggression were the main causes of aggressive behaviour towards people (43.1 and 39.6% of cases respectively). Most housesoiling problems involved urination (59%), followed by urination and defecation (32%) and defecation (9%), and the most common diagnosis was aversion to the litterbox (63.4%). Persian cats were presented more frequently for elimination problems than other breeds ($\chi^2 = 6.40$; $p < 0.01$). Cats with behaviour problems came from pet shops more frequently than control cats ($\chi^2 = 10.39$; $p < 0.001$). Cats without outdoor access showed significantly more behaviour problems than those with outdoor access ($\chi^2 = 31.85$; $p < 0.001$) and aggression problems were more common in single cat households than in multicat households ($\chi^2 = 25.47$; $p < 0.001$). Intact females were presented more frequently for aggression problems than neutered females ($\chi^2 = 7.74$; $p < 0.01$).

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1. Introduction

The number of cats has increased and even surpassed the number of dogs in some countries (PFMA, 2000; Kendall and Ley, 2006; APPMA, 2008). In fact, a recent study concluded that the majority of people prefer cats as pets (Casey and Bradshaw, 2008a) because they are easier to take care of than dogs and also because they appear to be more independent (Zasloff and Kidd, 1994; Downey and

Ellis, 2008). In Spain, around 13% of all households have one or more cats, and 6.3% of households have both dogs and cats (Veterindustria, 2006). The increase of cat numbers can partly explain why the number of cats referred for behaviour problems has increased over the last few years (Heath, 2007; Casey and Bradshaw, 2008b).

The term "behaviour problem" is used to describe any behaviour shown by an animal (in the context of this paper, a cat) that is unacceptable to the owner. Some behaviour problems such as sexual marking are normal behaviours, that is, they belong to the normal feline behaviour repertoire and yet, when shown in a domestic environment, became objectionable (Casey and Bradshaw, 2008b).

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Behaviour problems are important for several reasons. First, they are one of the most common reasons for relinquishment of cats (Kass, 2007). A 2004 report concluded that half of the adopted kittens had at least one behaviour problem after the first month (Wright and Amoss, 2004) and other authors have found that behaviour problems account for more than one third of all the unsuccessful rehoming of abandoned cats (Shore, 2005).

Second, behaviour problems are important from a public health point of view because although the percentage of aggression problems is higher towards other cats than towards people, a high number of aggressive episodes are still directed towards people. Physical and psychological consequences of cat attacks should not be underestimated, mainly when the target of the attacks are elderly people, children or people with compromised immunity (August, 1995; Heath, 2002).

Finally, but not less important, behaviour problems may negatively affect the welfare of the animals, mainly when they are a consequence of stress or anxiety (Heath, 2007).

Epidemiological studies are important to understand the risk factors of behaviour problems and thus the best preventive measures. However, results can vary depending on the geographical area and the source of data. For example, previous studies have shown that data obtained through a questionnaire to the general practitioners are different from those obtained through a consultation in a referral service (Fatjó et al., 2006). To the best of our knowledge, this is the first epidemiological study on behaviour problems in cats carried out in Spain and our objectives were to (1) describe the most common behaviour problems in a feline population attended in a referral behaviour service and (2) identify the risk factors on which preventive measures should be based.

2. Animals, materials and methods

2.1. Study side

The study group came from the caseload of the Animal Behaviour Service at the Veterinary Teaching Hospital of the Universitat Autònoma de Barcelona located 20 km from Barcelona. The Veterinary teaching Hospital attends around 13,000 cases per year, primarily referred by veterinarians from the Barcelona metropolitan area.

2.2. Data collection

Cases included in this retrospective study were presented for behaviour problems at the Animal Behaviour Service between 1998 and 2006. Data were obtained using a standard questionnaire that the owner had to fill at the time of consultation. Information obtained during the interview is summarized in Table 1. All cats included in the study underwent a thorough physical examination.

To categorize aggressive cases, four main criteria were used: the beginning and evolution of the problem, the target of aggression, the context in which aggression occurred and the cat body posture during the aggression. The classification of aggressive behaviour used in this

Table 1

Main areas in which questionnaire was divided.

General information	Name of the cat Breed Sex and neutering status Age Age of adoption Source of adoption
Physical and social environment Social behaviour Nutrition Elimination habits Daily activities Training	
Description of behaviour problem	Beginning of the problem Context in which problem occurs Frequency and evolution
Description of the cat's general behaviour	

paper is derived from that developed by other authors (Voith, 1983; Beaver, 1989a; Overall, 1997). Cases that did not fit any category were included in the group "other forms of aggression". If a cat showed more than one form of aggression, each form was considered and analyzed separately.

2.3. Control population

In order to identify risks factors, a control population was randomly selected from the general database of the same hospital during the same period of time. Control cats did not have any history of behaviour problems. Owners of control cats were interviewed using a standard questionnaire to obtain information on signalment, physical and social environment, and other relevant characteristics of cats.

2.4. Statistical analysis

The significance of the relationship between the occurrence of behaviour problems and cat characteristics was analyzed with a χ^2 -test. A value of $p < 0.05$ was considered significant for all analyses. All statistical analyses were performed using SPSS 15.0 for Windows.

For the analysis of breed predisposition, cats of the study group without a specific problem were included into the control group. Crossbreds were excluded from the analysis.

3. Results

A total of 336 cases presented from 1998 to 2006 were included in the study group. 178 (53%) were females (64.6% of which were castrated) and 158 (47%) were males (77.2% of which were castrated). The average age at evaluation was 4.5 years. 116 cats had been rescued from the street (45.8%), 74 were homebred (29.2%), 44 had been purchased from a petshop (17.4%), 14 had been obtained

from animal shelter (5.5%) and 5 had been purchased from a professional breeder (1.9%). The source of 83 cats (24.7%) was unknown. 42 cats (14%) had outdoor access, whereas 179 (59%) lived with other cats in the same household. 82 cats were pure bred cats (24%) and 254 (76%) were Domestic European Shorthair. Siamese and Persian accounted for the highest percentage of pure bred cats. The average age at which cats were adopted was 4.6 months (SD = 10.87).

Information was also collected on 189 control cats. 95 cats were males (75 of which were neutered; 78.95%) and 94 were female (68 of which were spayed; 72.34%). The average age at evaluation was 4.9 years. 87 cats were rescued from the street (46.52%), 58 (31.02%) were obtained from a private owner, 20 (10.70%) were acquired from an animal shelter, 13 (6.95%) were purchased from a petshop, and 9 (4.81%) were obtained from a professional breeder. 121 (64.64%) cats were indoor cats and 67 (35.64%) had outdoor access. 123 (65.78%) cats lived with other cats in the same household. 41 cats were purebred (21.69%) and 148 (78.31%) were Domestic European Shorthair. As in the study group, Persian (60.97%) and Siamese (26.83%) accounted for the highest percentage of pure bred cats. Mean age of cats at adoption was 3.9 months (SD = 32.95). Data of both groups are summarized in Table 2.

Aggression was the main owner complaints (171 diagnoses out of a total of 365; 47%), followed by inappropriate elimination (142/365; 39%), compulsive behaviour (13/365; 3.5%), excessive vocalization (10/365; 2.5%) and fear and phobias (9/365; 2.5%). Other problems such as anorexia, furniture scratching and over-activity were included in the same group (20/365; 5.4%) (Fig. 1). The average number of behaviour problems per cat was 1.1.

In relation to the target of aggression, cats showed aggression more frequently towards other cats (110/171; 64%) than towards people. When the aggression was directed towards people, the most common target was the owner (46/59; 78%).

Most cases of aggression directed towards family members met the criteria of play-related aggression (25/

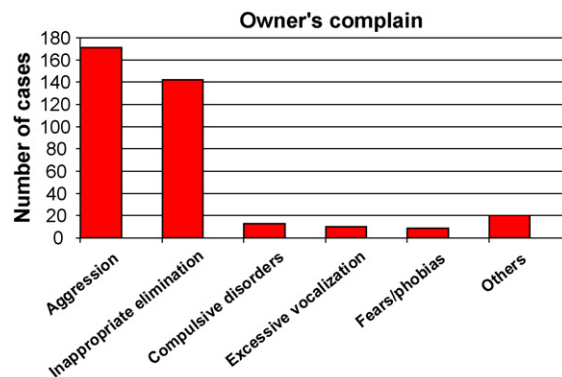


Fig. 1. Number of cases according to the owner complain.

58; 43.1%) and petting-related aggression (23/58; 39.6%). Almost twenty eight percent of cats showed at the same time play-related aggression and petting-related aggression. When the aggression was directed to unfamiliar people, 70% of cats showed defensive aggression (7/10), followed by offensive aggression (3/10; 30%). A similar result was found for intercat aggression and almost seventy percent showed defensive aggression (72/107; 67.3%) whereas 31.8% (34/107) showed offensive aggression towards another cat. In one case the owner was not able to describe the cat body posture.

Most of the elimination cases involved urination (84/142; 59%), followed by urination and defecation (45/142; 32%), and defecation (13/142; 9%). Although it seems to be a certain degree of overlap between some categories of diagnosis, the most frequent diagnosis was litterbox aversion (90/142; 63.4%) followed by urine marking (22/142; 15.5%; see Table 3). Persian cats showed elimination problems out the litterbox more frequently than other breeds ($\chi^2 = 6.40$; $p < 0.01$). No breed effect was found for any other behaviour problem.

The study population came more frequently from petshops than the control group ($\chi^2 = 10.39$; $p < 0.001$). The percentage of breeds did not differ between both groups.

A lower percentage of cats in the study group had outdoor access than in the control group (261/303; 86% in the study group vs. 121/188; 64.64% in the control group; $\chi^2 = 31.85$; $p < 0.001$). Cats from multicat households showed less aggressive problems towards family members than cats from households with only one cat ($\chi^2 = 25.47$; $p < 0.001$).

No effect of sex or neutering status was found on the likelihood of cats' showing behaviour problems except for the fact that intact females showed aggression problems more frequently than neutered ones ($\chi^2 = 7.74$; $p < 0.01$).

4. Discussion

According to the present study, aggression and elimination problems are the most commonly reported behaviour problems in urban Spanish cats. These results are similar to those obtained in other referral practices (Blackshaw, 1985; Borchelt and Voith, 1987; APBC, 2005;

Table 2
Characteristics of cats in the study and control group.

	Study group	Control group
N	336	189
Sex		
Females	178 (53%)	94 (49.74%)
Intact	63	26
Neutered	115	68
Males	158 (47%)	95 (50.26%)
Intact	36	20
Neutered	122	75
Source		
Street	116 (45.85%)	87 (46.52%)
Private owner	74 (29.25%)	58 (31.02%)
Pet shop	44 (17.39%)	13 (6.95%)
Animal shelter	14 (5.53%)	20 (10.70%)
Professional breeder	5 (1.98%)	9 (4.81%)
Mean age at evaluation (years)	4.5	4.9
Mean age of adoption (months)	4.6	3.9

Table 3
Distribution of diagnosis. Each cat may have more than 1 diagnosis.

Diagnosis	No. of cats (%)
Aggression	195 (49.2)
People-directed aggression	68 (17.2)
Defensive aggression towards family members	3 (0.7)
Play-related aggression	27 (6.8)
Petting-related aggression	26 (6.6)
Defensive aggression towards unfamiliar people	7 (1.8)
Offensive aggression towards unfamiliar people	3 (0.7)
Frustration related aggression	2 (0.5)
Animal-directed aggression	107 (27)
Defensive aggression towards family cat	72 (18.2)
Offensive aggression towards family cat	34 (8.6)
Defensive aggression towards dog family	1 (0.2)
Redirected aggression	20 (5)
House soiling	131 (33.1)
Inappropriate elimination related to the litter box	90 (22.7)
Marking	22 (5.5)
Others	19 (4.8)
Compulsive disorders	11 (2.8)
Psychogenic alopecia	6 (1.5)
Feline hyperaesthesia	4 (1)
Wool sucking	1 (0.2)
Ingestive disorders	2 (0.5)
Obesity	1 (0.2)
Pica	1 (0.2)
Fears/phobias	6 (1.5)
Towards other cats	3 (0.75)
Towards people	2 (0.5)
Towards sounds	1 (0.2)
Sexual behaviour	1 (0.2)
Miscellaneous	15 (3.8)
Others	35 (8.8)

Bamberger and Houpt, 2006). For instance, according to the Association of Pet behaviour Counsellors, around 39% of cats were referred for inappropriate elimination and 32% for aggression problems (APBC, 2005). These data indicate that these problems are very common and also that they may be particularly challenging and therefore poorly tolerated by many owners who will ask for advice more readily than when the cat shows another problem (Heath, 2007; Casey and Bradshaw, 2008b).

Depending on the source of information, there could be some discrepancies between epidemiologic studies. A previous study done in the same geographic area through a questionnaire sent to general practitioners concluded that the most common behaviour problem was inappropriate elimination followed by furniture scratching (Fatjó et al., 2006). According to the present study, however, furniture scratching accounted for a very small number of cases. This could be due to the fact that furniture scratching is not perceived as serious enough as to seek the advice of a behaviourist (Fatjó et al., 2006).

As other authors have reported, intercat aggression appears to be more frequent than aggression towards people (Houpt, 1998; Heath, 2002; APBC, 2005; Bamberger and

Houpt, 2006). In our study, the high proportion of multicat households (59%) combined with the high proportion of indoor cats (86%) could contribute to the high percentage of aggression between cats in the caseload (Heidenberg, 1997; APBC, 2005). Conflicts among cats may be more likely if the home range is small, as it occurs with indoor cats (Rochlitz, 2005). According to Leyhausen, indoor cats may show aggression as a result of dispersion being prevented. Furthermore, the lack of a developed communicative language in cats as compared with dogs, may increase the likelihood of aggressive encounters (Leyhausen, 1979). According to Bernstein and Strack, indoor cats should have access to at least two rooms and an additional space should be added in multicat households (Bernstein and Strack, 1996). In our study area, the proportion of indoor cats (study group 86%; control group 64.4%) was much higher than in other countries (Heidenberger, 1997; Levine et al., 2005). For instance, in UK, the majority of cats have access to the outside and less than 20% of cats were kept indoors permanently (APBC, 2005).

In the present study the percentage of indoor cats was higher in the study group than in the control group. These results suggest that behaviour problems may be associated with keeping cats indoors, as other authors have reported before (Mertens and Schär, 1988; Heidenberg, 1997; Buffington, 2002). It is important to emphasize, however, that quality of space is at least as important as the amount of space per se. Environmental enrichment in the form of vertical structures, scratching posts, and hiding places, for example, can improve the welfare of cats (Carlstead et al., 1993). Further, it is also very important to avoid either a monotonous and impoverished environment or unpredictable changes, as both could cause stress (Broom and Johnson, 1993; Rochlitz, 2005).

Aggression towards family members accounted for the majority of aggression cases towards people. According to our results and to previous studies, housing a single cat is correlated with a high likelihood of aggressive behaviour towards the owners (APBC, 2005). Cats kept without contact with other cats can direct play behaviour towards their owner more frequently than group-housed cats (Borchelt and Voith, 1987; Beaver, 2004) and play-related aggression was one of the most common causes of aggression. Indeed, it has been suggested that play-related aggression may be reduced by adding a second cat (Beaver, 2003a). The other most frequent form of aggression towards owners was petting-related aggression and a similar result has been found in other studies (Chapman, 1991). This could be due to the owners' inability to detect the subtle warning signals that precede an episode of petting-related aggression (Crowell-Davis et al., 1997).

Elimination problems related to the litter box, also categorized as a litterbox aversion, accounted for the great majority of housesoiling problems, as reported elsewhere (Beaver, 1989b; Halip et al., 1998; Bamberger and Houpt, 2006). Lack of appropriate management by the owners may explain the high incidence of litterbox aversion (Bamberger and Houpt, 2006).

Several studies have found a breed predisposition, with Persian and Siameses being more likely than other breeds to have elimination and aggression problems respectively

(Beaver, 1989a,b; Bamberger and Houpt, 2006). In the present study, Persian cats were presented more often for elimination problems than other breeds. Beaver (2003b) suggested that Persian cats might have difficulties in learning elimination habits.

The results of the present study suggest that the origin of the cat could be a risk factor and cats purchased from petshops were presented more often for behaviour problems than those adopted from other sources. Serpell and Jagoe (1995) found that dogs with dominance-type aggression and social fear came more frequently from pet shops than control dogs but to the best of our knowledge this is the first study reporting a similar effect on cat behaviour. There are at least two possible mechanisms that could explain this finding. First, it is widely accepted that cats should be exposed to a variety of stimuli during the socialization period between 2 and 7 weeks of age (Karsh and Turner, 1998; Mc Cune et al., 1995) and it is likely that many pet shops provide a relatively poor environment compared with other sources (Casey and Bradshaw, 2008a).

Secondly, early weaning may have long-lasting effects on behaviour. For example, early-weaned kittens may have difficulties to control the strength of their bite when playing (Overall, 1994a,b). Further, it seems that licking by the mother has beneficial effects on the stress response and, those individuals that have been subjected to intense licking shortly after birth are better able to adjust their stress response to the intensity of the stressor compared with animals that have been subjected to less licking and these effects also persist over time, as indicated in rats (Champagne et al., 2003). In addition, previous studies conducted in laboratory animals suggested that exposure to stressors during the early postnatal period or the post-weaning juvenile phase can permanently alter the neural system and consequently some aspects of the behaviour due to changes on the epigenome (Meaney, 2001; Champagne et al., 2003; Holmes et al., 2005). Therefore, inadequate breeding management could be a source of stress with long lasting effects. Whether cats from petshops are more likely than cats from other sources to be exposed to stressors early in life warrants further investigation.

In the majority of species, males are more aggressive than females (Nelson, 2000). Nevertheless, in some species such as golden hamster (*Mesocricetus auratus*), for example, females are more aggressive than males (Nelson, 2000). Several studies have reported that female cats are presented for aggression more frequently than male cats (Wright, 1990; Palacios et al., 2007). In our study, no significant differences were found for aggression problems between males and females. Nevertheless, intact females were more aggressive than neutered females. This result is at variance with those obtained in dogs which concluded that neutering can increase aggressive behaviour in females due to the elimination of progesterone, which has a calming effect (Hart and Eckstein, 1997). In contrast, removal of progesterone seems to have the opposite effect in female golden hamsters (Payne and Swanson, 1972). One study with feral cats by Neville and Remfry (1984) concluded that neutering reduced aggressive behaviour in

both sexes (Neville and Remfry, 1984), although the underlying physiological mechanisms are not known. We did not analyse the effect of neutering of females on each category of aggression. However, the possibility that entire females may be more territorial than neutered as they would need to defend a larger territory to raise offspring can not be ruled out.

5. Conclusion

According to the present study five risk factors were identified. Cats purchased from petshops and those without outdoor access had a higher risk of developing behaviour problems; intact females showed more frequently aggression problems than neutered females, the likelihood of aggression towards people was significantly higher in single cat households, and, Persian cats were presented for elimination problems more frequently than other breeds. The knowledge of these risk factors could help to advise owners and prevent behaviour problems.

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